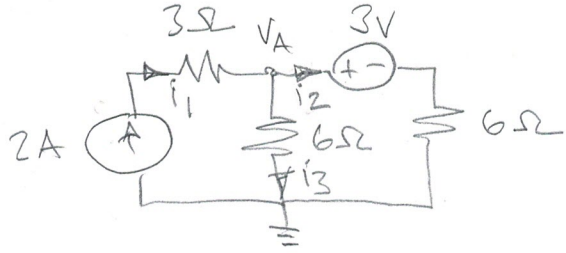
EOOBE Lösningsförslag

230817

1)




NODANALYS - ANSÄTT NODSP. $V_A$ OCH STRÖMMARNA KSTL I NOD A GER

$$i_1 - i_2 - i_3 = 0 \quad (1)$$

MEN $i_3 = \frac{V_A}{6}$ (2)

$$i_2 = \frac{V_A - 3}{6} \quad (3)$$

$$i_1 = 2\,A \quad (4)$$

(2)-(4) i (1) GER

$$2 - \frac{V_A - 3}{6} - \frac{V_A}{6} = 0$$

$$2 + \frac{3}{6} = \frac{1}{3} V_A$$

$$3\frac{15}{6} = V_A$$

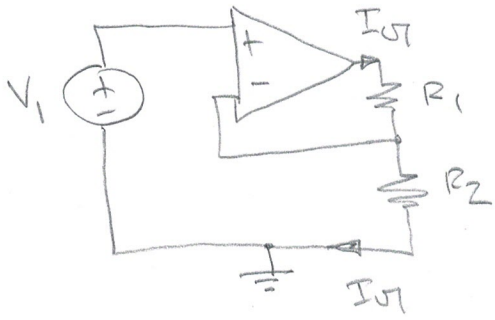
$$V_A = 7.5 \text{ VOLT}$$

STRÖMMEN I BLIR

$$i = i_3 = \frac{V_A}{6} = \frac{7.5}{6} \simeq 1.25A$$

SVAR: $i = 1.25A$

2)




KSPL över R

$-V_1 + R_2 I_{UT} = 0$

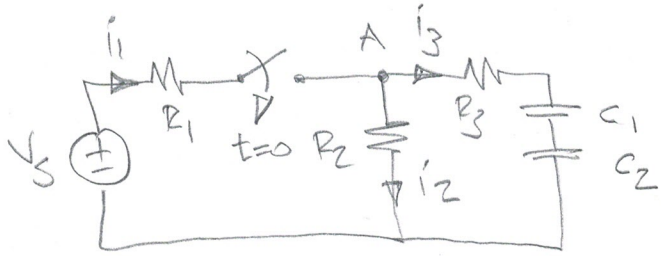
$I_{UT} = \frac{V_1}{R_2}$

SVAR: $I_{UT} = \frac{V_1}{R_2}$

(EN VOLT-TILL-STRÖM OMVANDLARE)

(FÖR ENKEL UPPGIFT!)

3)




$R_1 = 3\,k\Omega$

$R_2 = 6\,k\Omega$

$R_3 = 2\,k\Omega$

$C_1 = 12\,nF$

$C_2 = 36\,nF$

$V_s = 9\,Volt$

BRYTAREN STÄNGS VID $t=0$, KOND OLADDADE.

FÖR NOD A GÄLLER $i_1 - i_2 - i_3 = 0$ (1)

MEN $i_2 = \frac{V_A}{R_2}$ och (2)

$$V_A - R_3 i_3 - V_{C_1} - V_{C_2} = 0 \quad (3)$$

DÄR $\begin{cases} V_{C_1} = V_{C_1}(0) + \frac{1}{C_1}\int i_3\,dt = \frac{1}{C_1}\int i_3\,dt \\ V_{C_2} = V_{C_2}(0) + \frac{1}{C_2}\int i_3\,dt = \frac{1}{C_2}\int i_3\,dt \end{cases}$ (4)

(4) i (3) ger

$$V_A - R_3 i_3 - \underbrace{\left(\frac{1}{C_1}+\frac{1}{C_2}\right)}_{=\frac{1}{C_E}}\int i_3\,dt = 0 \quad (5)$$

DERIVERA MAP t

$$\frac{di_3}{dt} + \frac{1}{R_3 C_E} i_3 = 0$$

SOM HAR LÖSNING $i_3(t) = i_3(0)\,e^{-t/R_3 C_E}$

$i_3(0)$ DÅ? VID $t=0$ HAR KOND INGEN LADDNING

DÅ GÄLLER (1) & (3) $\times$ (2)

$$(3) \Rightarrow V_A - R_3 i_3(0) = 0 \quad (6)$$

$$(2) \Rightarrow i_2(0) = \frac{V_A}{R_2}$$

$$\Rightarrow i_1(0) = V_A\left(\frac{1}{R_3}+\frac{1}{R_2}\right)$$

och $V_s - R_1 i_1(0) = V_A \Rightarrow i_1(0) = \frac{V_s - V_A}{R_1}$

$$\Rightarrow V_A\left(\frac{1}{R_3}+\frac{1}{R_2}\right) = \frac{V_s - V_A}{R_1}$$

▷

$$V_A\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)=\frac{V_S}{R_1}$$

$$V_A(0)=V_S\frac{\frac{1}{R_1}}{\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}}=9\frac{\frac{1}{3}}{\frac{1}{3}+\frac{1}{6}+\frac{1}{2}}=3\ \text{VOLT}$$

MED (6) FÅS $i_3(0)=\frac{V_A(0)}{R_3}=\frac{3}{2\cdot10^3}=1.5\cdot10^{-3}\ A$

STRØMMEN $i_3(t)=1.5\cdot10^{-3}e^{-t/0.018}=1.5\cdot10^{-3}e^{-55.6t}$ A.

SPÆNNINGEN OVER $C_2$ BLIR ENLIGT (4)

$$v_{C_2}(t)=v_{C_2}(0)+\frac{1}{C_2}\int i_3(t)\,dt=$$

($v_{C_2}(0)=0$)

$$=\frac{i_3(0)}{C_2}\int_0^t e^{-t/R_3C_E}\,dt=$$

$$=\frac{i_3(0)}{C_2}\left[-R_3C_E e^{-t/R_3C_E}\right]_0^t=$$

$$=\frac{1}{C_2}i_3(0)R_3C_E\left(1-e^{-t/R_3C_E}\right)$$

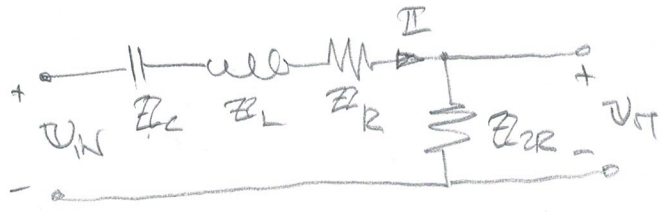
MED SIFFRER

$$v_{C_2}(t)=0.75\left(1-e^{-55.6t}\right)\ \text{VOLT}$$

SVAR: a) $i_3(t)=1.5\cdot10^{-3}e^{-55.6t}$ A

b) $v_{C_2}(t)=0.75\left(1-e^{-55.6t}\right)$ V

4)




KOMPLEX REPR. BERÄKNA $H(\omega)$

KSPL ger

$$\begin{cases} -V_{IN} + (Z_C + Z_L + Z_R) I = V_{UT} & (1) \\ -V_{UT} + Z_{2R} I = 0 \Rightarrow I = \dfrac{V_{UT}}{Z_{2R}} & (2) \end{cases}$$

(2) i (1)

$$-V_{IN} + (Z_C + Z_L + Z_R)\frac{V_{UT}}{Z_{2R}} = V_{UT}$$

$$V_{IN} = (Z_C + Z_L + Z_R)\frac{V_{UT}}{Z_{2R}} + V_{UT} =$$

$$= V_{UT}\left[Z_C + Z_L + Z_R + Z_{2R}\right]\frac{1}{Z_{2R}}$$

B.U.R.

$$H(\omega) = \frac{V_{UT}}{V_{IN}} = \frac{Z_{2R}}{Z_C + Z_L + Z_R + Z_{2R}} =$$

$$= \frac{2R}{\frac{1}{j\omega C} + j\omega L + R + 2R} =$$

$$= \frac{2j\omega RC}{1 - \omega^2 LC + 3j\omega RC}$$

STUVA OM LITE

$$H(\omega) = \frac{2j\omega RC\left[1 - \omega^2 LC - 3j\omega RC\right]}{(1-\omega^2 LC)^2 + (3\omega RC)^2} =$$

$$= \frac{2j\omega RC(1-\omega^2 LC) - 6(\omega RC)^2}{(1-\omega^2 LC)^2 + (\omega RC)^2}$$

OM $V_{IN}$ ÄR I FAS MED $V_{UT}$ SÅ MÅSTE IM-DELEN AV $H(\omega) = 0$, DVS

$$1 - \omega_0^2 LC = 0$$

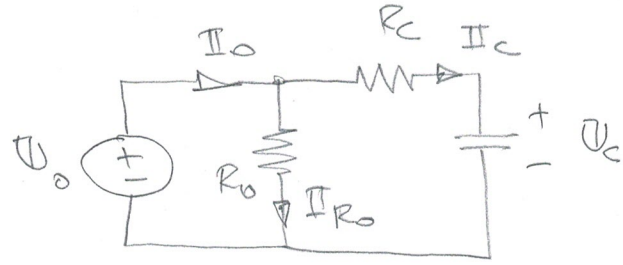
$$\omega_0 = \frac{1}{\sqrt{LC}}$$

SVAR:

a) $H(\omega) = \dfrac{2j\omega RC}{1-\omega^2 LC - 3j\omega RC}$

b) $\omega_0 = \dfrac{1}{\sqrt{LC}}$

5)




BESTÄM AMPLITUDEN PÅ STRÖMMEN $I_0$

$$\begin{cases} f = 5.3\ \text{KHz} \\ R_0 = 3\ \text{K}\Omega \\ R_C = 2\ \text{K}\Omega \\ C = 10\ \text{nF} \end{cases}$$

VI VET ATT $V_C = 3\angle 0^\circ$ V

FÖR NOD A GÄLLER

$$I_0 - I_{R_0} - I_C = 0 \qquad (1)$$

DÄR $I_{R_0} = \frac{V_0}{R_0}$ (2)

KSPL för högersida: $V_C + R_C I_C = V_0$ (3)

MEN $V_C = Z_C I_C = \frac{1}{j\omega C} I_C \Rightarrow I_C = j\omega C V_C$ (4)

(5)

(4) i (3) $V_C + R_C j\omega C V_C = V_0$ (6)

(5) i (2) $I_{R_0} = \frac{1}{R_0}(1 + R_C j\omega C) V_C$

(6) & (4) i (1)

$$I_0 - \frac{1}{R_0}(V_C + R_C j\omega C V_C) - j\omega C V_C = 0$$

$$I_0 = \left[\frac{1}{R_0}(1 + R_C j\omega C) + j\omega C\right] V_C$$

$$= \left[\frac{1}{R_0} + j\omega C\left(\frac{R_C}{R_0} + 1\right)\right] V_C$$

SIFFROR

$$I_0 = \left[\frac{1}{3\cdot 10^3} + j\,2\pi\,5300\cdot 10\cdot 10^{-9}\left(\frac{2}{3} + 1\right)\right] 3\angle 0^\circ =$$

$$= 1\cdot 10^{-3} - j\,1.67\cdot 10^{-3} = 1.94\cdot 10^{-3}\angle -59.1^\circ$$

SVAR: $|I_0| \simeq 1.94$ MA